EXERCISE 07

Feedback and Root Locus

- We are halfway through the semester!

$\Rightarrow$ Appropriately this means we will now enter stage 2 of the course.

What we have been analyzing

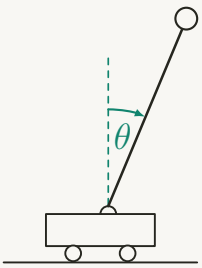
Control Systems

What we will look at now

- The halfpoint is a good opportunity for a “what happened so far”...

RECAP

○ Modelling



- How to describe a system with physics.
- Represent in this form:

$$\dot{x}(t) = f(x(t), u(t))$$
$$y(t) = g(x(t), u(t))$$

- We discussed the difference between
  - Inputs: $u(t)$
  - States: $x(t)$
  - Outputs: $y(t)$

○ Classification & Linearization

- We looked at these classifications:
  - Linear / Nonlinear
  - Causal / Non-causal
  - Time-variant / Time-invariant
  - Dynamic / Static
- In this course all systems are **Linear & Time-invariant (LTI)**.

- We can say this, because systems can be **linearized**.

$$\begin{aligned}\dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= g(x(t), u(t))\end{aligned}$$

Jacobi Matrices

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

⇒ Good approximation around **equilibrium point** (x_e, u_e)

○ Time response & Stability

- Here we solved the differential equation.

$$y(t) = \underbrace{Ce^{At}x_0}_{\text{Initial condition response}} + \underbrace{C\int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau}_{\text{Forced response}} + Du(t)$$

- And saw that a system is asymptotically stable if all

$$\text{Re}\big(\text{eigenvalues}(A)\big) < 0.$$

○ Transfer Functions

- We can transform our systems into frequency-domain by using the Laplace-transform.

$$G(s) = C(s\mathbb{I} - A)^{-1}B + D$$

- Convolutions
- Integrals
- Derivatives

} **simple multiplications**

- We looked at our new description of this realm:

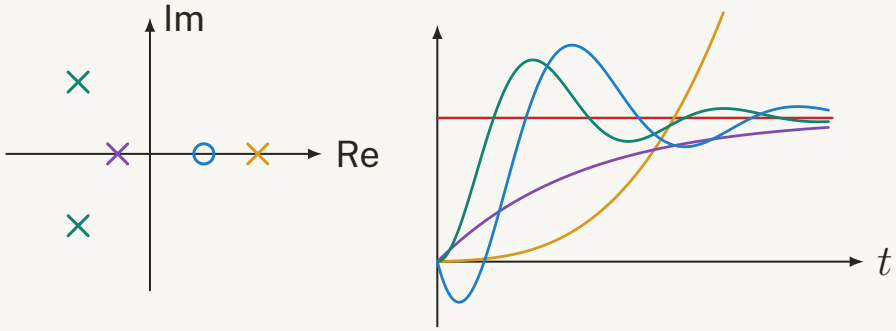
$\Rightarrow$ Complex-Frequency $s = \sigma + j\omega$

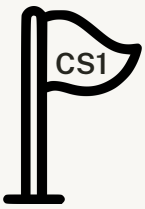
○ Poles & Zeros

- We can characterize a system's behaviour by looking at its poles and zeros:

$$G(s) = k\,\frac{(s - z_1)(s - z_2)\cdots(s - z_m)}{(s - p_1)(s - p_2)\cdots(s - p_n)}$$

- The poles & zeros tell us how a system responds



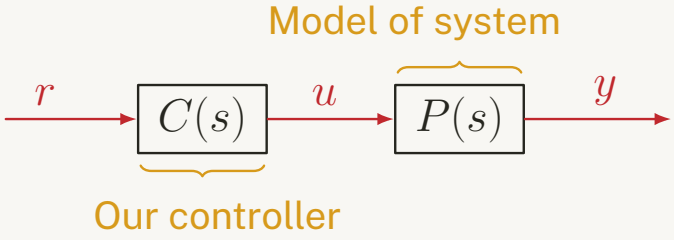


Checkpoint!

Now we go back, to how it all began...

1 | Feedback

The initial problem



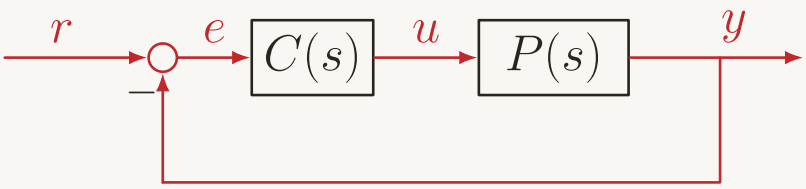
r: reference
u: input
y: output

- This we now call the **open-loop** $L(s)$

$$L(s) = C(s)P(s)$$

- Now, let's finally introduce **feedback**:

e: error



- The closed loop transfer function we derived like this:

$$y = C \cdot P \cdot e\,, \quad e = r - y$$

$$y = CP(r - y) \;\Rightarrow\; y(1 + CP) = CPr$$

$$y = \frac{CP}{1 + CP}\, r$$

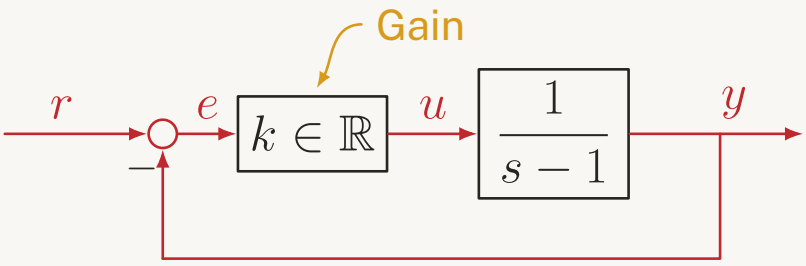
- We call this tf from $r \to y$ $T(s)$:

$$T(s) = \frac{C(s)P(s)}{1 + C(s)P(s)} = \frac{L(s)}{1 + L(s)}$$

- $T(s)$ is also called **complementary sensitivity function**.

The essential question is now, how does feedback affect our system?

EX. Effect of proportional control

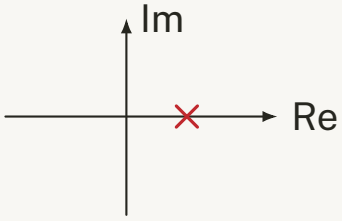


- Without feedback, the open-loop is:

$$L(s) = \frac{k}{s-1}$$

$\Rightarrow$ **Unstable**, no matter our gain!!

Pole-Zero-Plot



- If we now close the loop, the tf becomes:

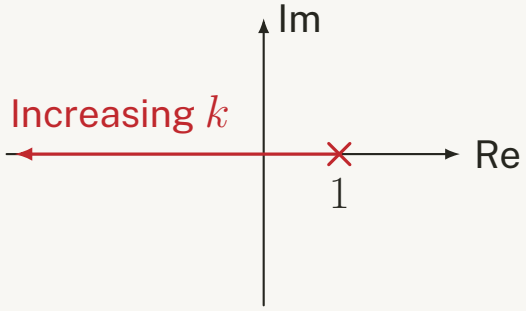
$$T(s) = \frac{L(s)}{1+L(s)} = \frac{\frac{k}{s-1}\cdot(s-1)}{1+\frac{k}{s-1}\cdot(s-1)} = \frac{k}{s+k-1}$$

Where are the poles now?

- denominator $\stackrel{!}{=} 0$

$$s_p + k - 1 = 0 \quad \Rightarrow \quad s_p = 1 - k$$

$\Rightarrow$ It depends on k!



In this case, the closed loop becomes stable for $k \in (1, \infty)$

This is our next important plot...

2 | The root-locus plot

A plot that shows the position of poles and zeros over different gains k, is called a:

Root-Locus Plot

- To draw a root locus plot given a plant $P(s)$ you must follow a set of rules...

CHEAT SHEET

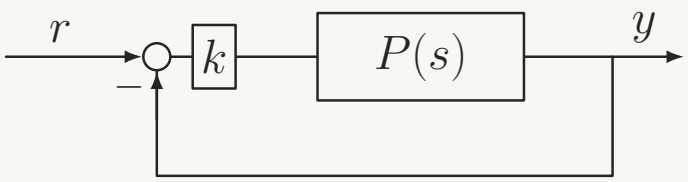
Root-Locus Sketching Rules

Rule	Description
Symmetry	Closed-loop poles are symmetric about the real axis
# of branches	# closed-loop poles = # open-loop poles
Start ($k \to 0$)	Closed-loop poles start at open-loop poles
End ($k \to \infty$)	If $\deg N = \deg D$: poles $\to$ zeros If $\deg N < \deg D$: excess poles $\to \infty$
Real-axis locus	For $k > 0$, locus exists to the left of an odd number of poles/zeros
Asymptotes	$\theta_q = \frac{(2q+1)180^\circ}{n-m}, \quad q = 0, \ldots, n-m-1$ $s_{\text{com}} = \frac{\sum \text{Re}(p_i) - \sum \text{Re}(z_i)}{n-m}$

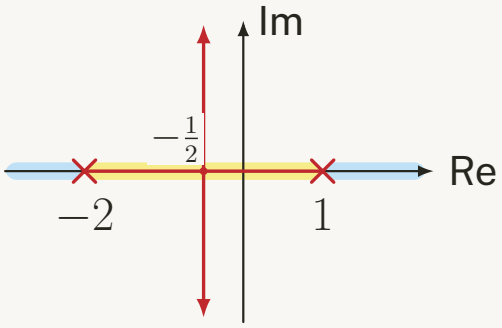
Let's ignore these for a second and establish the rules visually...

Some examples:

Example 1: two poles



$$P(s) = \frac{1}{(s+2)(s-1)} = \frac{1}{s^2+s-2}$$



▶ **Start ($k \to 0$) & # of branches:** Draw the open-loop **poles & zeros**.

Every pole starts one **branch**: here 2 branches.

▶ **Real-axis locus:** Identify existing / non-existing sections on the real axis:

for $k > 0$ the locus lies left of an **odd** number of real poles/zeros.

- Trick: from $+\infty$ to the first real pole/zero: **No line**.

  After that, **alternate**.

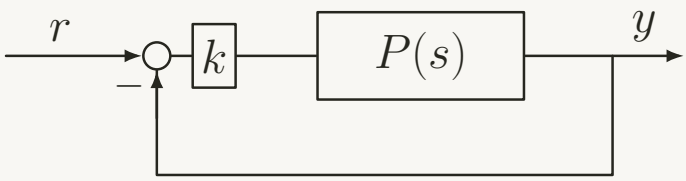
(Only poles/zeros on the real axis count. A double pole/zero does not switch.)

▶ **Asymptotes (preview):** The poles meet at $s = -\frac{1}{2}$ and leave the real axis at $90°$.

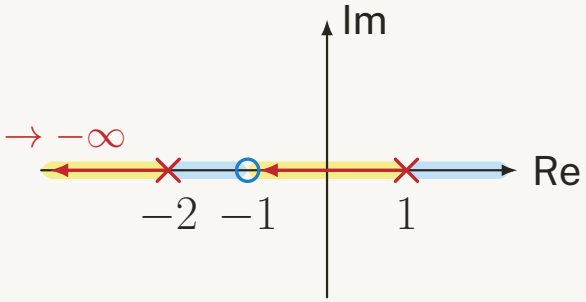
(Here this is exactly s_{com}, see Example 3.)

Ok, but what if we add zeros?

Example 2: adding a zero



$$P(s) = \frac{s+1}{(s+2)(s-1)} = \frac{s+1}{s^2+s-2}$$



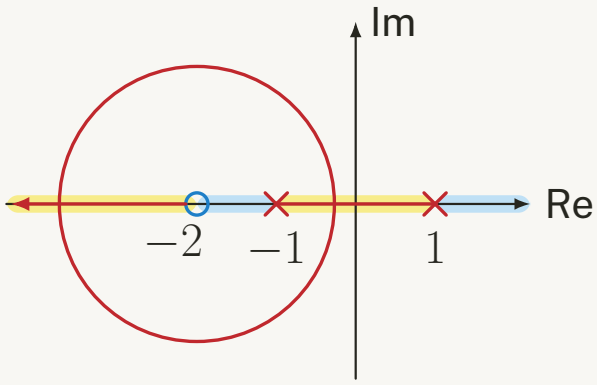
▶ **End ($k \to \infty$):** Zeros **do not move**! Poles go to **zeros**.

If $\deg N < \deg D$, the **excess** poles go to ∞.

Here: pole $1 \to$ zero -1, pole $-2 \to -\infty$.

Watch what happens if we swap the left pole and zero:

$$P(s) = \frac{s+2}{(s+1)(s-1)} = \frac{s+2}{s^2-1}$$



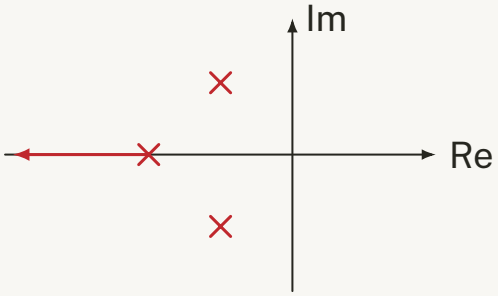
▶ **Symmetry:** Locus is **symmetric** about the **real axis.**

End again: one pole → zero -2, the other → $-\infty$.

Example 3: more poles than zeros

- The last rules are unfortunately harder...

$$P(s) = \frac{1}{(s+2)(s+1+j)(s+1-j)} = \frac{1}{s^3+4s^2+6s+4}$$



▶ **Asymptotes:** If $n > m$, the excess poles go along **asymptotes.**

The asymptotes cross the real axis at their centre s_{com} and leave it at angles θ_q:

$$\theta_q = \frac{(2q+1)\cdot 180^\circ}{n-m}, \qquad s_{\text{com}} = \frac{\sum \text{Re}(p_i) - \sum \text{Re}(z_i)}{n-m}$$

$$q = 0, 1, \ldots, (n-m-1), \quad n \mathrel{\widehat{=}} \#\text{poles}, \quad m \mathrel{\widehat{=}} \#\text{zeros}$$

Here:

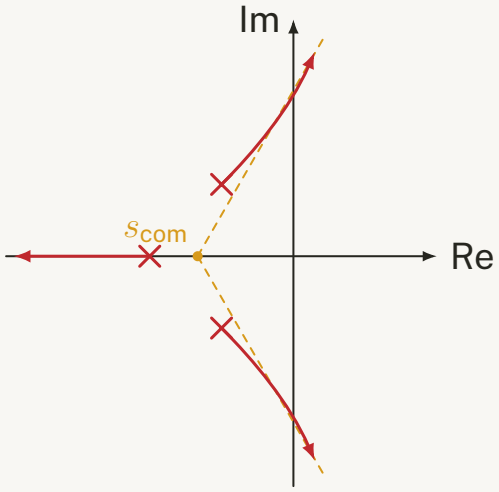
$$\theta_0 = \frac{180^\circ}{3-0} = 60^\circ, \qquad \theta_1 = \frac{3\cdot 180^\circ}{3-0} = 180^\circ$$

$$\theta_2 = \frac{5\cdot 180^\circ}{3-0} = 300^\circ = -60^\circ$$

$$s_{\text{com}} = \frac{(-2)+(-1)+(-1)}{3-0} = -\tfrac{4}{3}$$

Check Example 1: $n-m=2 \Rightarrow \theta = 90^\circ,\ 270^\circ,\ s_{\text{com}} = \dfrac{-2+1}{2} = -\tfrac{1}{2}$ ✓

The break-away point equals s_{com} only in such simple cases (two poles, no zeros) – in general it does not.

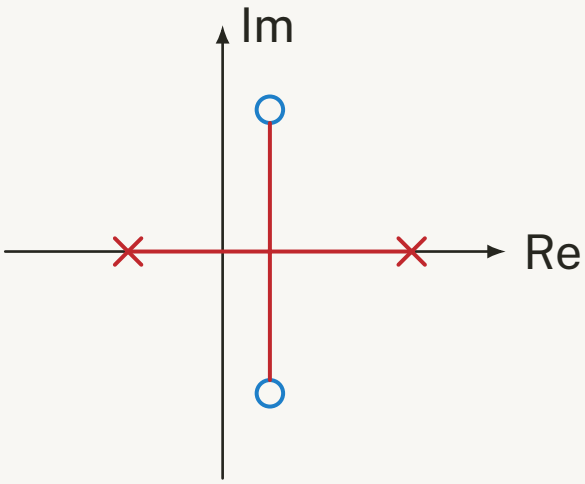


Intermediate quiz!!

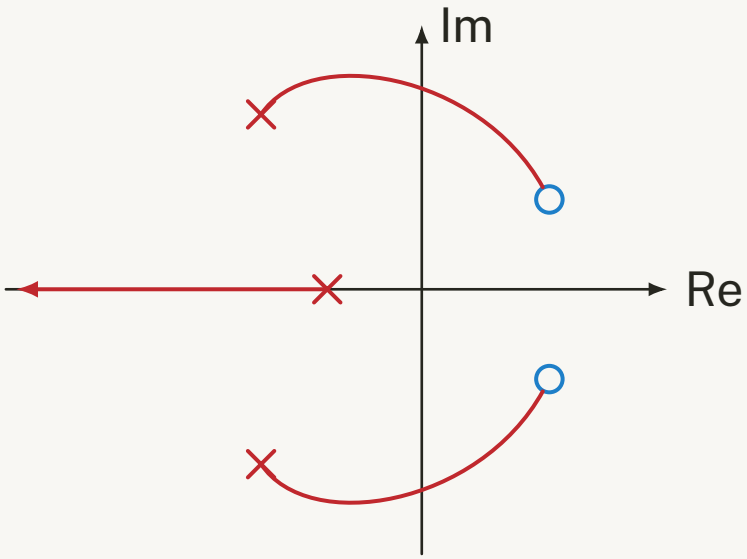
QUIZ

Draw the root loci for these plots:

Q1)

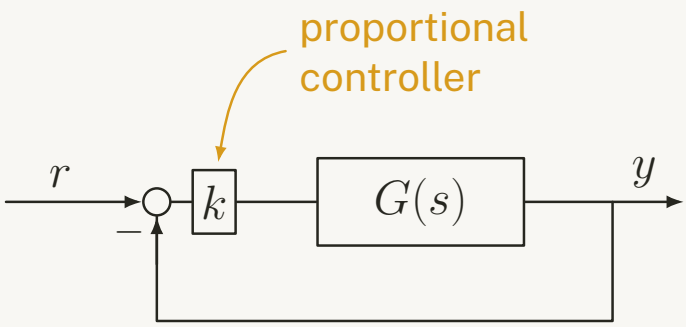


Q2)



3 | Understanding the root locus

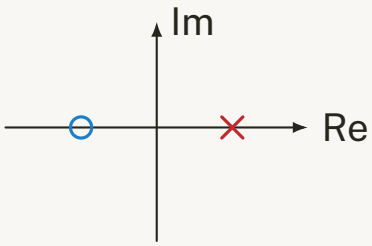
- So only by multiplying the signal that goes into the plant with an amplification (or gain) k, we have a lot of control over the system's behaviour!



How can we understand the root locus intuitively?

EX. Consider the following system:

$$G(s) = \frac{(s+1)}{(s-1)}$$



- Now let's add a proportional controller k and look at the closed loop.
- TF of closed loop:

$$T(s) = \frac{L(s)}{1+L(s)} = \frac{k \cdot \dfrac{(s+1)}{(s-1)}}{1 + k \cdot \dfrac{(s+1)}{(s-1)}}$$

$$= \frac{k \cdot \dfrac{(s+1)}{(s-1)}}{\dfrac{(s-1)+k(s+1)}{(s-1)}} = \frac{k(s+1)}{(s-1)+k(s+1)}$$

- Analyzing this shows us why the plotting rules work:

The zeros stay in the **same spot** $\forall k$

$$\frac{k\,(s+1)}{(s-1)+k\,(s+1)}$$

Increasing k means the open-loop zero becomes a more important closed-loop **pole**.

Closed-loop pole: $(s-1)+k\,(s+1)=0 \;\Rightarrow\; s_p(k)=\dfrac{1-k}{1+k}$

$k=0:\; s_p=+1$ (open-loop pole) $\qquad k\to\infty:\; s_p\to -1$ (open-loop zero)

$\Rightarrow$ This is why the closed-loop poles move towards the open-loop **zeros**!

Now we know everything to tackle root-locus exam questions!

Exam question

Problem: Consider the closed-loop system shown in Figure 11, where L represents a linear time-invariant system and $k \in \mathbb{R}$. The root locus of L for $k > 0$ is shown in Figure 12.

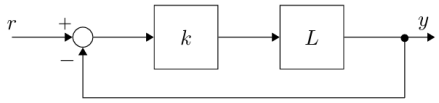


Figure 11: Standard feedback architecture. System L and proportional gain k.

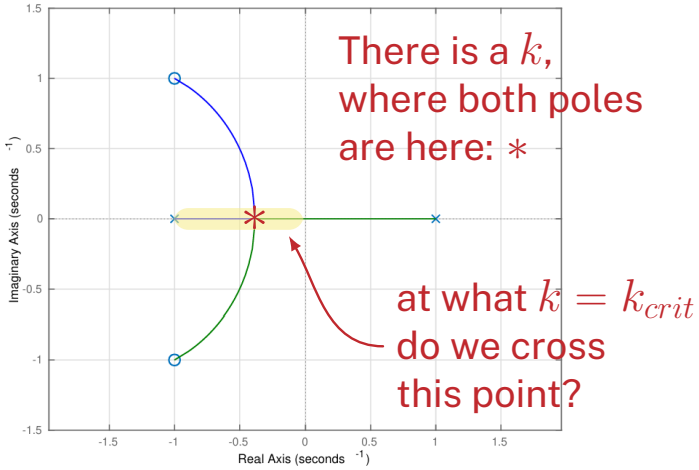


Figure 12: Root Locus of L. Poles are represented by $\times$ symbols, zeros by o symbols.

Q24 **(0.5 Points)** Mark the correct answer.
The open-loop system L is asymptotically stable.

[A] True [X] False

Q25 **(0.75 Points)** Mark the correct answer.
There exist two gains, k_1 and k_2, with $0 < k_1 < k_2$, such that for all $k \in [k_1, k_2]$, the closed-loop system is stable, and the closed-loop step response does not exhibit any overshoot.

[X] True [B] False

SOLUTION

Q24) Condition for asymptotic stability:

$\Rightarrow$ All poles < 0.

Here: $p_1 = -1,\ p_2 = 1$ $\leftarrow$ open-loop unstable!!

Q25) Condition for stability:

$\Rightarrow$ All poles ≤ 0

Condition for no overshoot:

$\Rightarrow$ All poles on **real axis** $(\in \mathbb{R})$

So we must evaluate, if there is a range of k, where both poles are in this realm.

$\Rightarrow$ We use **Start** and **Real-axis locus**:

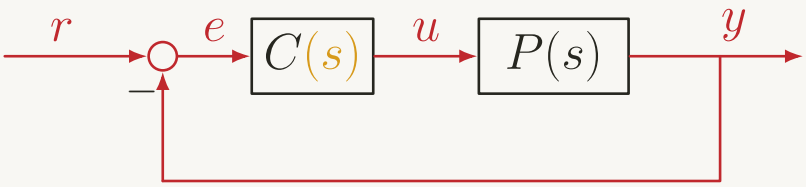
Both poles first move towards each other on the real axis and break away at $s \approx -0.38 < 0$ (read off the plot).

Just before break-away, both poles are real and negative $\Rightarrow$ **True**.

Now we understand the first (and most complex) of three main plots we will look at.

4 | Dynamic controllers: PID

- We see now, how powerful a simple scalar as a controller can be. But most of the time, our controller will be a dynamic system itself!



- A typical $C(s)$ design is combining these three components
  - Proportional action: k_P
  - Integral action: $k_I \cdot \frac{1}{s}$
  - Derivative action: $k_D \cdot s$

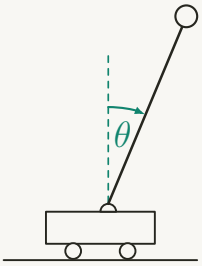
$\Rightarrow$ This is called a **PID-controller**

$$C_{PID}(s) = k_P + k_I \cdot \frac{1}{s} + k_D \cdot s$$

- Now we choose our gains k_P, k_I, k_D, and write the total system $L(s)$ as a function of just one gain k_{rl} by bringing $L(s)$ into root-locus-form:

$$L(s) = \frac{k_{rl}}{s^q} \cdot \frac{(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_{n-q})}$$

Example: inverted pendulum



Linearized EoM:

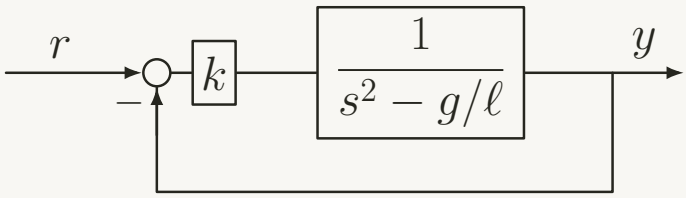
$$m\ell^2\ddot{\theta} = mg\ell\,\theta + u$$

↓ Laplace (zero initial conditions)

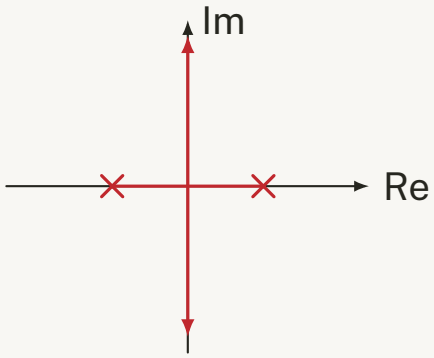
$$P(s) = \frac{\Theta(s)}{U(s)} = \frac{1}{m\ell^2 s^2 - mg\ell} = \frac{1}{m\ell^2} \cdot \frac{1}{s^2 - g/\ell}$$

The constant prefactor is absorbed into the gain k, so below we drop it.

Let's try and add a p-controller



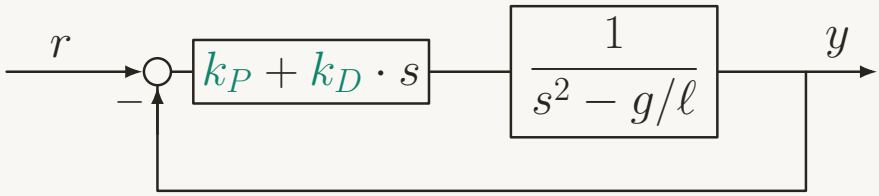
- To find out if we can make this system asymptotically stable, we draw the root locus:



Poles will stay on **Im-axis.**

⇒ **Never asymptotically stable...**

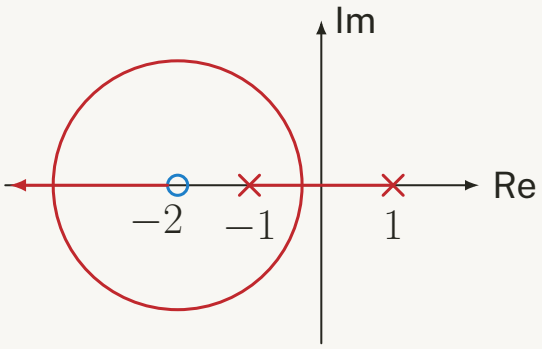
What if we choose a PD-controller?



We bring $L(s)$ into rl-form:

$$L(s) = \frac{k_P + k_D s}{s^2 - g/\ell} = \underbrace{k_D}_{k_{rl}} \cdot \frac{s + k_P/k_D}{\left(s - \sqrt{g/\ell}\right)\left(s + \sqrt{g/\ell}\right)}$$

Drawing root-locus again:



Now we can **stabilize** the system!!

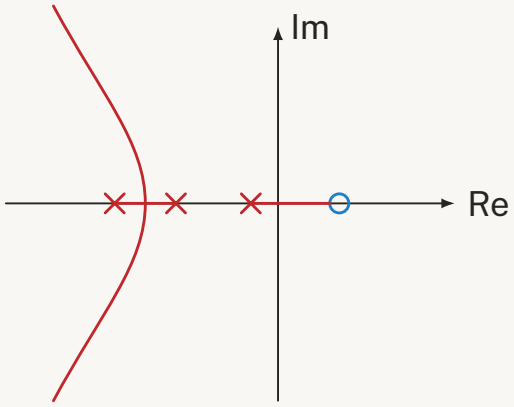
(here drawn for $\sqrt{g/\ell} = 1$, $k_P/k_D = 2$)

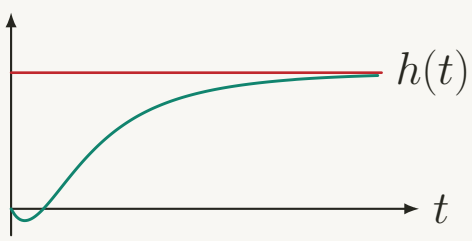
Almost done!

$\Rightarrow$ Quiz question!

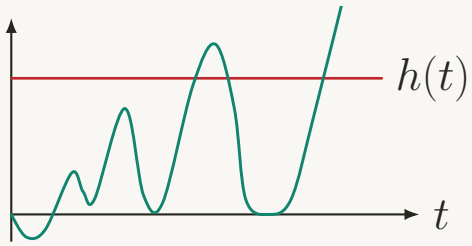
QUIZ

Q3) Given the open-loop pole-zero plot, find the closed loop step responses for a very small, and a very large proportional gain. (hard)





$k \mathrel{\widehat{=}}$ small



$k \mathrel{\widehat{=}}$ large

